Test Data

$z_1 = \begin{pmatrix} 30 \\ 15 \end{pmatrix}$ y=60 $z_2 = \begin{pmatrix} 40 \\ 20 \end{pmatrix}$ y=80 $z_3 = \begin{pmatrix} 50 \\ 25 \end{pmatrix}$ y=100

Outlier X_s	NO Regularizer with (as if λ=0)	Ridge Regularizer with λ=3	Ridge Regularizer with λ=10
$\begin{pmatrix} 11 \\ 3 \end{pmatrix}$ 22	W = 0, 2, 0 ||W||²=4	W = 0.1, 1.95, 0.03 ||W||²=3.8	W = 0.13, 1.87, 0.09 ||W||²=3.5
$\begin{pmatrix} 30 \\ 3 \end{pmatrix}$ 22	W = 2.7, 0.67, 0.64 ||W||²=8.11	W = 1.2, 0.72, 0.77 ||W||²=2.45	W = 0.54, 0.73, 0.81 ||W||²=1.48
$\begin{pmatrix} 60 \\ 3 \end{pmatrix}$ 22	W = 4, 0.29, 0.78 ||W||²=16.7	W = 1.77, 0.32, 0.99 ||W||²=4.2	W = 0.82, 0.33, 1.1 ||W||²=1.92

NO Regularizer with (as if λ=0)

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	60	80	100

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	32.4	42.3	52.3

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	24.4	31.2	38

Ridge Regularizer with λ=3

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	59	78.7	98.4

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	34.1	45.1	56.1

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	26.2	34.4	42.5

Ridge Regularizer with λ=10

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	57.4	76.5	95.6

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	34.6	46	57.4

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	26.8	35.4	44.1